October-5.

Day-6

Ethernet LAN Switching (Part 2).

1 hexadecimal character = 4 bit
2 hexadecimal characters = 8 bit (1 byte)

IPV4 ethernet type = 0x0800
IPV6 ethernet type = 0x86DD

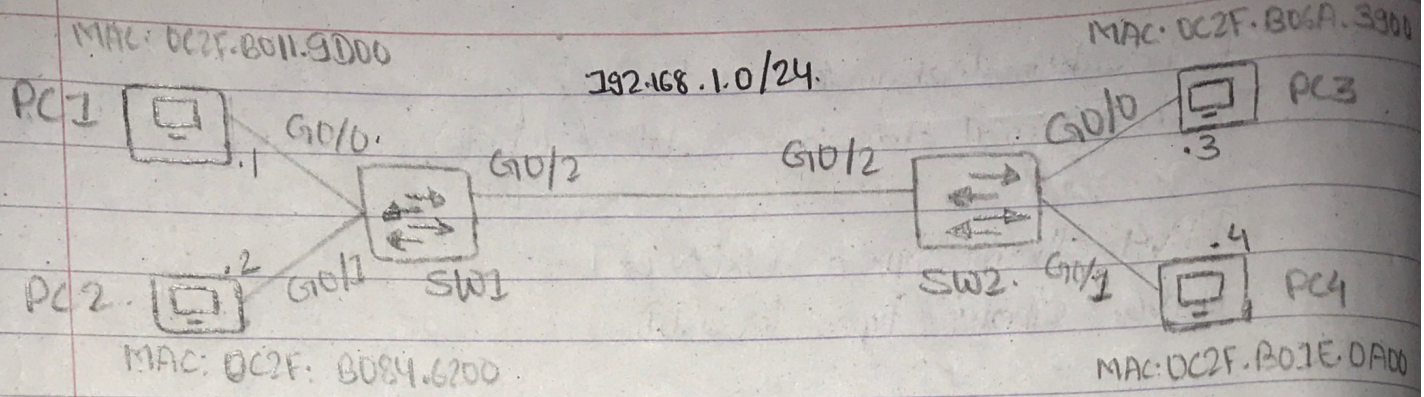
Ethernet Frames

- The preamble + SFD is usually not considered part of Ethernet header, although it is sent with every Ethernet frame.

- Therefore, the size of Ethernet header + trailer is 18 bytes (6+6+4+2). (Src, Dst, Type, FCS) (Type=2, FCS=4).

- The minimum size for an Ethernet frame (Header + Payload [Packet] + Trailer) is 64 bytes & max-size is 1518 byte.
- 64 bytes - 18 bytes (header + trailer) = 46 byte.
- Therefore the minimum payload (packet) size is 46 bytes.
- If the payload is less than 46 bytes, padding bytes are added, and these bytes are all 0s.

- For e.g if we send 34 byte packet, 12-byte of padding will be added. ie 34+12 = 46 byte



SW1 Mac-Add Table

	MAC	Interface
1.	.9D00	G0/0

SW-2 MacAdd Table.

	MAC	Interface
1	.9D00	G0/2
2.	.3900	G0/0

When a device sends some data to another device, it doesn't just include a source & destination MAC. Encapsulated with that frame is an IP packet. The IP packet consists of source & destination IP address.

For eg.

1). If PC1 wants to send data to PC3, the source IP will be 192.168.1.1 & destination IP will be 192.168.1.3. The src mac o MAC will be .9D00, however PC1 doesn't actually know PC3's MAC address. When you send data to another computer ig, you enter IP addr. not MAC address. So, user entered IP address 192.168.1.3 as destination, but PC1 has to discover PC3's MAC address by itself.

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The switches are layer 2 devices, they don't operate at layer 3, so they need to use MAC addd. not IP add.

So, PC1 wants to send Ethernet frame to PC3, but first it has to learn PC3's MAC address

To do so, it uses ARP (Address Resolution Protocol).

For 1
Src IP: 192.168.1.1
Dst IP: 192.168.1.3
Src MAC: 9D00
Dst MAC: ???

Address Resolution Protocol (ARP).

- It is used to discover layer 2 address (MAC address) of a known layer 3 address (IP address).

For eg, ~~our~~ in our example PC1 knowns PC3's layer 3 address (IP addr) but it doesn't know its layer 2 (MAC) address.

The ARP ~~con~~ process consists of two messages:

i) ARP Request - sent by the device that wants to know MAC of other device.
ii) ARP Reply - sent to inform the ~~M~~ requesting device of MAC address.

In our eg - PC1 sent ARP req, & PC3 sent ARP reply.

ARP request is sent as broadcast ethernet frame.

b ARP request is broadcast = sent to all hosts on network.

beos layer 2 address of destination host is unknown, it broadcast the request & wait for reply from current device.

ARP reply is unicast.

ARP request = broadcast (sent to all ~~de~~ host on network)
ARP reply = unicast

To comple 1, first PC1 has to send ARP request which is as below.

FFFF.FFFF.FFFF = broadcast MAC add.

ARP Request
Src IP: 192.168.1.1
Dst IP: 192.168.1.3
Src MAC: 0C2F.B011.9D00
Dst MAC: FFFF.FFFF.FFFF

So, PC1 has prepared ARP request to be sent. It sends frame out its network interface and it is received by SW1. SW1 then adds PC1 MAC to its table. Now, SW1 broadcast it through all interfaces except from where it was received This is similar to what Switch does with Unknown Unicast Frame.

SW1 sends out via G0/1 & G0/2. ~~G0/1~~, PC2 receives via G0/1 but ignores the frame as destination IP address doesn't match PC2 Mac Address.

SW2 now learns PC1 MAC address & adds it to its table associating it with G0/2 interface.

Now, SW2 also sends out frame via all interfaces except G0/2 to find destination MAC address. PC4 ignores frame.

PC3 recognize destination IP matches its own IP address. Now, it sends ARP reply.

Though ARP req. mssg was a broadcast mssg, bcoz PC1's MAC address was used as Source MAC, of ARP req, PC3 now knows PC1's MAC & can send ARP reply to PC1 without having to broadcast the frame

ARP Reply
Src IP: 192.168.1.3
Dst IP: 192.168.1.1
Src MAC: 0C2F.B06A.3908
Dst MAC: 0C2F.B011.9D00

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ICMP - Internet Control Message Protocol

PC3 sends ARP reply frame out of its interface, SW2 receiveds it. It enters PC3 MAC addres on its table. Now ARP reply is unicast & SW2 knows MAC of PC1, so it forwards it to PC1 through SW2.

ARP Table.

- Use arp -a to view ARP table (Windows, macOS, Linux)
  It has 3 columns:
  - Internet Address - IP address (Layer 3 address)
  - Physical Address - MAC address (Layer 2 address)
  - Type static: - default entry = static
    learned via ARP = dynamic.

Ping

- A network utility that is used to test reachability.
- Measures round-trip time.

  Eg: From PC1 → PC3 and back from PC3 → PC1.
- Uses two message.

i) ICMP echo Request
ii) ICMP echo Reply.

PC's won't broadcast ICMP echo req, it is sent to specific host. For this MAC add. of destination is required. So, ARP must be used first.

The command for ping is: ping ip-address

```
PC1 #
1 PC1 # ping 192.168.1.3
To Sending 5,100-byte ICMP echos to 192.168.1.3, timeout is
    2 seconds: .!!!!
Success rate is 80% (4/5), round trip min/avg/max = 20/20/22 ms
PC1 #
```

By default, a ping in CISCO IOS sends 5 ICMP echo requests, and then you should get 5 ICMP echo replies back. The default size of each ping is 100-byte.

The period (.) indicates failed ping & (!) exclamation indicates succesful ping.

The first ping failed because of ARP

In CISCO, IOS, to see arp-table use the command.

Show arp (from priv. exec mode.

MAC address table.

Show mac address-table

It has 4 cols:

VLAN -

MAC Address -

TYPE -

Ports/Interface -

The dynamic MAC address are removed from MAC address table after 5 mins of inactivity. This is known as aging

We can also manually remove MAC add. from table.

(1) clear mac address-table dynamic

It clears out all dynamic MAC address.

(2) clear mac address-table dynamic address 0c2f.b011.9f00

It removes given dynamic MAC add. from table.

(3) clear mac address-table dynamic interface Gi0/0

Clears dynamic mace addres from table whose interface is Gi0/0

QUIZ

Q.1) You send a 36-byte ping to another computer & perform a packet capture to analyze the network traffic. You notice a long series of bytes of 000000000..... at the end of Ethernet payload. How can you explain these zeroes.?

a) Pings are a series of 0s b) They are padding bytes c) They're Ethernet FCS

Q.2) Which of the message is sent to all hosts on local network?

a) ARP request (b) ARP reply (c) ICMP echo req. d) ICMP echo reply

Q.3) Which fields are present in the output of the show mac address-table command on Cisco ~~router~~ switch?

a) MAC, ports b) VLAN, MAC, ports c) VLAN, MAC, Type, Ports.

d). Internet Address, Physical Add., Type.

Q4) Which type of frame does a switch send out of all interfaces, except the one the frame was received on?

a) Broadcast, unknown unicast b) Broadcast, known unicast

c) Known unicast, unknown unicast d) Broadcast, unknown uni, known unicast.

Q5) Which command is used on CISCO switch to clear all dynamic MAC address on a specific interface from MAC address table.

a) clear mac address-table interface interface-id.

b) clear mac-address-table dynamic interface interface-id

c) clear mac-address table dynamic interface interface-id.

d) clear mac address-table dynamic interface interface-id.

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